Name
Date
Course

The Law of Sines (Part II): The Ambiguous Case (SSA) Homework

Find the missing side lengths and angle measures of each triangle. In other words, solve each triangle.

1. $A = 101°$, $a = 3$ mm, $b = 2$ mm

2. $C = 132°$, $c = 9$ ft, $b = 11$ ft

③ $B = 80°$, $b = 7.5$ in, $a = 17$ in

④ $A = 52°$, $c = 8.6$ km, $a = 6.9$ km

(5) $C = 29°$, $a = 77\text{cm}$, $c = 22\text{cm}$

(6) $A = 114°$, $b = 15\text{m}$, $a = 14\text{m}$

7. $B = 75^\circ$, $b = 20$ mi, $c = 12$ mi

8. $C = 21^\circ$, $b = 6$ ft, $c = 2.7$ ft

Homework Answers

① $B \approx 40.876°$

$C \approx 38.124°$

$c \approx 1.887$ mm

One Triangle Possible

② $132° + 65.269° \not< 180°$

$197.269° \not< 180°$

No Triangle Possible

③ $\sin A \not\approx 2.232$

No Triangle Possible

④

$C \approx 79.16°$	$C \approx 100.839°$
$B \approx 48.839°$	$B \approx 27.16°$
$b \approx 6.592$ km	$b \approx 3.997$ km

Two Triangles Possible

⑤ $\sin A \not\approx 1.6968$

No Triangle Possible

⑥ $78.18077° + 114° \not< 180°$

$192.18° \not< 180°$

No Triangle Possible

⑦ $C \approx 35.419°$

$A \approx 69.58°$

$a \approx 19.4$ mi

One Triangle Possible

⑧

$B \approx 52.785°$	$B \approx 127.215°$
$A \approx 106.215°$	$A \approx 31.785°$
$a \approx 7.234$ ft	$a \approx 3.9685$ ft

Two Triangles Possible